Derivatives and Alternative Investments

Exam Formula-Check Notes

Coverage: Lecture 1, Lecture 2 sections 1–3 only, Lecture 4. Lecture 3 is excluded.

Notation

Symbol	Meaning
S_t	Spot price at time t
S_T	Spot price at maturity/expiration
F_t	Forward or futures price at time t
F_0	Initial forward/futures price
K	Delivery price or option strike
T	Time to maturity, in years
r	Continuously compounded risk-free rate
q	Continuous income/dividend yield
I	Present value of known cash income
U	Present value of known storage cost
u	Proportional storage-cost rate
y	Convenience yield
c, p	European call and put values
C, P	American call and put values
N	Number of futures contracts
Q	Contract size
h	Hedge ratio

Check: Most pricing formulas below assume continuous compounding unless stated otherwise.

1. Payoffs and Contract Values

Derivative Basics

A derivative payoff can be written generally as

$$\boxed{\text{payoff at } T = g(X_T)}$$

where X_T is the underlying variable at maturity.

Payoff is the terminal cash flow. **Profit** is payoff minus initial cost and financing. **Value** is the current market value of the contract.

Forwards

Long forward: agree to buy at K.

$$\boxed{\text{long forward payoff} = S_T - K}$$

Short forward: agree to sell at K.

$$\boxed{\text{short forward payoff} = K - S_T}$$

For a non-income-paying investment asset:

$$\boxed{F_0 = S_0 e^{rT}}$$

With simple annual interest, use the matching simple-compounding version, e.g.

$$F_0 = S_0(1 + rT).$$

Check: If quoted $F_0 > S_0 e^{rT}$, cash-and-carry: borrow, buy asset, short forward. If quoted $F_0 < S_0 e^{rT}$, reverse cash-and-carry: short asset, invest proceeds, long forward.

Futures Daily Settlement

For N contracts and contract size Q:

$$\boxed{\text{P\&L}_t^{\text{long}} = NQ(F_t - F_{t-1})}$$

$$\boxed{\text{P\&L}_t^{\text{short}} = NQ(F_{t-1} - F_t)}$$

Check: Margin is collateral, not the price of the futures contract. Daily settlement reduces credit risk but can create liquidity risk.

Options

Call option payoff:

$$\boxed{\max(S_T - K, 0) = (S_T - K)^+}$$

Put option payoff:

$$\boxed{\max(K - S_T, 0) = (K - S_T)^+}$$

Profit, ignoring interest in payoff diagrams:

$$\text{long option profit} = \text{payoff} - \text{premium}.$$

American exercise weakly increases value:

$$\boxed{C \geq c, \qquad P \geq p}$$

2. Hedging With Futures

Choosing Hedge Direction

- **Short hedge:** use when the firm owns, produces, or expects to receive an asset it will later sell.
- **Long hedge:** use when the firm expects to buy an asset later and wants protection against rising prices.
- **Cross hedge:** use a related futures contract when no liquid futures contract exists on the exact exposure.

Simple contract count when hedge ratio is one:

$$\boxed{N = \frac{\text{quantity to hedge}}{\text{contract size}}}$$

Basis Risk

Basis is spot minus futures:

$$\boxed{b_t = S_t - F_t}$$

For a short hedge opened at $t = 1$ and closed at $t = 2$:

$$\text{futures gain} = F_1 - F_2,$$

$$\boxed{\text{effective sale price} = S_2 + (F_1 - F_2) = F_1 + b_2}$$

For a long hedge opened at $t = 1$ and closed at $t = 2$:

$$\text{futures gain} = F_2 - F_1,$$

$$\boxed{\text{effective purchase price} = S_2 - (F_2 - F_1) = F_1 + b_2}$$

Check: If $b_2 = 0$, the hedge locks in F_1. Otherwise, b_2 is the remaining basis risk.

Cross-hedge basis decomposition:

$$\boxed{S_2 - F_2 = (S_2 - S_2^*) + (S_2^* - F_2)}$$

The first term is mismatch between physical assets; the second is the usual basis for the futures underlying.

Minimum-Variance Hedge Ratio

For a short hedge using hedge ratio h:

$$\Delta V = \Delta S - h\Delta F.$$

Residual variance:

$$\mathrm{Var}(\Delta V) = \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F).$$

Optimal hedge ratio:

$$\boxed{h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \rho\frac{\sigma_S}{\sigma_F}}$$

Contract count:

$$\boxed{N^* = h^*\frac{Q_A}{Q_F} \qquad \text{or} \qquad N^* = h^*\frac{V_A}{V_F}}$$

Check: If the regression uses price changes, use physical quantities. If it uses percentage returns, use values.

3. Forward and Futures Pricing

Investment Assets

Non-income-paying asset:

$$\boxed{F_0 = S_0 e^{rT}}$$

Known cash income with present value I:

$$\boxed{F_0 = (S_0 - I)e^{rT}}$$

Known continuous yield q:

$$\boxed{F_0 = S_0 e^{(r-q)T}}$$

Value of an existing long forward with delivery price K:

$$\boxed{f = (F_0 - K)e^{-rT}}$$

For an existing short forward:

$$f_{\text{short}} = (K - F_0)e^{-rT}.$$

Commodity Pricing

Fixed storage cost with present value U:

$$\boxed{F_0 = (S_0 + U)e^{rT}}$$

Proportional storage cost u:

$$\boxed{F_0 = S_0 e^{(r+u)T}}$$

With proportional storage cost u and convenience yield y:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

For consumption commodities, arbitrage may give only an upper bound:

$$\boxed{F_0 \le (S_0 + U)e^{rT}}$$

Contango: futures prices rise with maturity. **Backwardation:** futures prices fall with maturity.

Futures Prices and Expected Spot Prices

The futures price is not just a forecast:

$$\boxed{F_0 = \mathrm{E}(S_T)e^{(r-k)T}}$$

where k is the required expected return on the underlying asset.

Systematic risk	Relation	Implication
None	$k = r$	$F_0 = \mathrm{E}(S_T)$
Positive	$k > r$	$F_0 < \mathrm{E}(S_T)$
Negative	$k < r$	$F_0 > \mathrm{E}(S_T)$

Check: Do not confuse backwardation/contango as curve shapes with normal backwardation/contango as statements about expected future spot prices.

4. Options and Moneyness

Four Basic Positions

Position	Payoff	Exposure
Long call	$\max(S_T - K, 0)$	Upside
Short call	$-\max(S_T - K, 0)$	Upside loss
Long put	$\max(K - S_T, 0)$	Downside protection
Short put	$-\max(K - S_T, 0)$	Downside loss

Break-Even Points

Ignoring interest:

$$\text{long call break-even} = K + \text{premium},$$

$$\text{long put break-even} = K - \text{premium}.$$

Moneyness

	Call	Put
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$

Intrinsic value:

$$\boxed{\text{call intrinsic value} = \max(S - K, 0)}$$

$$\boxed{\text{put intrinsic value} = \max(K - S, 0)}$$

Time value:

$$\boxed{\text{time value} = \text{option price} - \text{intrinsic value}}$$

A standard U.S. exchange-traded stock option contract usually covers 100 shares, and quoted option prices are usually per share.

5. Option Price Restrictions

Inputs to Stock Option Prices

Input increases	Call	Put
S_0	Increases	Decreases
K	Decreases	Increases
σ	Increases	Increases
r	Usually increases	Usually decreases
Dividends	Decreases	Increases

Upper Bounds

$$\boxed{c \le S_0, \qquad C \le S_0}$$

$$\boxed{p \le K, \qquad P \le K}$$

For a European put:

$$\boxed{p \le Ke^{-rT}}$$

Lower Bounds

For European options on a non-dividend-paying stock:

$$\boxed{c \ge \max(S_0 - Ke^{-rT}, 0)}$$

$$\boxed{p \ge \max(Ke^{-rT} - S_0, 0)}$$

6. Put-Call Parity

For European options on a non-dividend-paying stock:

$$\boxed{c + Ke^{-rT} = p + S_0}$$

Equivalent form:

$$\boxed{c - p = S_0 - Ke^{-rT}}$$

Fiduciary call:

$$\text{call} + \text{cash } Ke^{-rT}.$$

Protective put:

$$\text{put} + \text{stock}.$$

Both pay $\max(S_T, K)$ at maturity.

Parity Arbitrage Rule

Compare the two packages:

$$\underbrace{c + Ke^{-rT}}_{\text{fiduciary call}} \quad \text{and} \quad \underbrace{p + S_0}_{\text{protective put}} \quad .$$

Mispricing	Arbitrage direction
$c + Ke^{-rT} > p + S_0$	Sell fiduciary call; buy protective put
$c + Ke^{-rT} < p + S_0$	Buy fiduciary call; sell protective put

Check: Buy the cheap package and sell the expensive package. The maturity payoffs cancel; the mispricing is cash today.

7. Binomial Trees

One-Step Setup

Stock movement:

$$\boxed{S_u = S_0 u, \qquad S_d = S_0 d, \qquad u > 1, \quad d < 1}$$

Terminal option values are f_u and f_d.

Delta and Replication

Make Δ shares minus one option riskless:

$$S_0 u\Delta - f_u = S_0 d\Delta - f_d.$$

Thus:

$$\boxed{\Delta = \frac{f_u - f_d}{S_0 u - S_0 d} = \frac{f_u - f_d}{S_0(u - d)} = \frac{f_u - f_d}{S_u - S_d}}$$

Check: Delta is local to the current node and time step. In a multi-step tree it changes across nodes and must be rebalanced.

Risk-Neutral Probability and Price

One-step option price:

$$\boxed{f = e^{-rT}\left[pf_u + (1-p)f_d\right]}$$

Risk-neutral probability:

$$\boxed{p = \frac{e^{rT} - d}{u - d}}$$

Equivalent condition defining p:

$$\boxed{pS_0u + (1-p)S_0d = S_0e^{rT}}$$

Check: p is not generally the real-world probability of an up move. It is the pricing probability that makes the stock earn the risk-free rate.

Multi-Step Backward Induction

If option life is T and there are n steps:

$$\boxed{\Delta t = \frac{T}{n}}$$

At each node:

$$\boxed{f = e^{-r\Delta t}\left[pf_u + (1-p)f_d\right]}$$

For two steps:

$$f_u = e^{-r\Delta t}\left[pf_{uu} + (1-p)f_{ud}\right],$$

$$f_d = e^{-r\Delta t}\left[pf_{ud} + (1-p)f_{dd}\right],$$

$$f = e^{-r\Delta t}\left[pf_u + (1-p)f_d\right].$$

Two-step terminal stock prices:

$$\boxed{S_0u^2, \qquad S_0ud, \qquad S_0d^2}$$

Choosing Tree Parameters

For volatility σ and time step Δt:

$$\boxed{u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}}$$

Then:

$$\boxed{p = \frac{e^{r\Delta t} - d}{u - d}}$$

Check: Given S_0, u, d, r and the payoff, the option price is pinned down by arbitrage. The stock's real-world expected return does not enter.